

Task 1 - Selection and Research

Jurisdiction-Aware Credit Decision Governance Monitor

Selected entity	Revolut
Domain	AI-assisted credit decisioning / credit scoring fairness / AI governance
Jurisdictions	United States and European Union
Format	Option C with a quantitative component and a lightweight evidence-output prototype

1. Selection and Rationale

This project selects Revolut as the named regulated entity and focuses on AI-assisted credit decisioning and credit scoring fairness across the United States and the European Union. Revolut is a suitable case because it presents itself as a global digital financial institution and has public US credit product disclosures. The assignment treats the US use case carefully as Revolut-operated or Revolut-partnered credit products, rather than assuming Revolut is always the final legal creditor.

The choice is intentionally realistic but bounded. Revolut is a digital-first financial services group with a public ambition to operate globally. It has launched or supported credit-related products in multiple markets, but the legal entity, issuer, partner-bank structure and regulatory perimeter may differ by product and jurisdiction. That makes Revolut useful for this assignment because the compliance function must coordinate governance evidence even when product ownership and legal responsibility are not identical across countries.

Revolut is also a better fit for this project than a traditional global bank. A traditional bank may operate locally ring-fenced systems, local data stores and separate model governance processes by national subsidiary. A digital-first financial institution is more likely to operate comparable product, customer onboarding and model-monitoring workflows across markets, even when the legal creditor or partner-bank arrangement differs. The project therefore does not assume that Revolut uses one global credit model or one pooled customer dataset. It studies a more realistic governance problem: how a cross-border digital financial institution can maintain comparable credit decisioning controls while allowing jurisdiction-specific data rights, evidence requirements and escalation routes.

The selected domain is AI-assisted credit decisioning / credit scoring fairness. This domain is appropriate for a RegTech assignment because it combines model risk, consumer impact, explainability, fairness monitoring and jurisdiction-specific compliance evidence. Credit decisions affect access to financial services, and automated or semi-automated decisioning can scale both useful consistency and harmful error. A governance tool must therefore do more than report model accuracy. It must show how evidence is produced, which data can be accessed, which human owner is responsible and what happens when the tool detects a weak explanation, a proxy discrimination signal or an incomplete high-risk AI governance record.

2. Regulatory Divergence: US vs EU

The US and EU do not merely ask for the same model report in different formats. The US side is anchored in adverse action explanation and fair lending review. The EU side treats creditworthiness or credit scoring AI as a high-risk use case and therefore asks for lifecycle governance, data governance, technical documentation, human oversight, record-keeping and post-deployment monitoring. That divergence drives the tool architecture.

The political and regulatory difference is not simply technical. The US framework places weight on individualised accountability after a credit decision: if a person is denied credit or treated unfavourably, the creditor must be able to identify the principal reasons in a specific and usable form. The CFPB has made clear that use of complex algorithms does not excuse failure to provide specific and accurate adverse action reasons. The EU framework places greater weight on institutional lifecycle governance: if an AI system is used for creditworthiness assessment or credit scoring, the system falls into a high-risk governance setting because the use case can affect access to important economic services. These are not mutually exclusive values, but they produce different product requirements.

A US module that only reports aggregate bias metrics would miss the adverse-action problem. A rejected applicant needs a reason that can be understood and challenged; a SHAP value or internal model contribution is not enough on its own. Conversely, an EU module that only generates individual reason codes would miss the lifecycle governance problem. A high-risk AI system needs evidence of risk management, data governance, human oversight, logging and monitoring before and after deployment.

The table below translates the regulatory divergence into concrete tool design implications.

Theme	United States	European Union	Design implication
Primary regulatory question	Can the creditor provide specific, accurate reasons for adverse action and detect fair lending risk?	Can the provider or deployer evidence high-risk AI governance before and after deployment?	The tool needs different calculation modules, not just different report templates.
Sensitive-data posture	US live path hard-drops direct protected attributes and relies on proxy-risk signals or separately approved privileged review.	EU audit path may use sensitive audit fields only in a segregated, lawful, access-controlled environment for bias testing.	Shared governance schema, not shared sensitive-data pipeline.
Explanation posture	SHAP-style attribution is internal evidence only and must be mapped to approved reason codes.	Explanation is part of broader high-risk AI transparency, documentation and oversight evidence.	The same model score can generate different compliance evidence.
Lifecycle posture	Main tool output is an adverse action evidence package and fair lending review trigger.	Main tool output is a pre-deployment evidence gate, audit bias testing, drift monitoring and human oversight workflow.	Jurisdiction affects workflow timing, not only reporting language.

3. Why This Needs a Jurisdiction-Aware Tool

A memo or spreadsheet can describe the regimes, but it cannot reliably enforce data-path separation, version rule changes, produce consistent evidence packages or escalate failures to the right owner. The prototype therefore uses a `jurisdiction_config` layer, synthetic data and evidence outputs to show how a compliance team would see different risk signals under US and EU settings.

The main design risk is building a decorative jurisdiction selector. To avoid that, the prototype makes the jurisdiction choice operational. US data is physically stripped of audit-only protected fields before analysis. EU audit data keeps sensitive audit fields only in a controlled testing path. The US module produces reason-code and proxy-risk evidence. The EU module produces high-risk AI governance, bias testing and drift evidence. The output schema is shared, but the calculations are not.

This is the central product thesis of the assignment: shared governance schema, not shared sensitive-data pipeline. The tool shares evidence fields such as jurisdiction, model version, rule version, risk signal, owner, required action, deadline and audit log reference. It does not share sensitive data access, computation modules or human review triggers across jurisdictions. This matters because RegTech tools are not neutral dashboards. They translate legislative and political choices into data permissions, workflow timing and accountability structures.

4. Synthetic Data and Scope Boundary

The quantitative component uses synthetic data because real Revolut customer credit data is not available. Synthetic data is suitable for this assignment because the purpose is not to estimate Revolut's real-world disparate impact; it is to demonstrate how a jurisdiction-aware tool would route data, compute metrics, produce evidence and trigger review. The figures should therefore be read as method demonstration, not as empirical findings about Revolut or its customers.

The scope boundary is deliberate. The project does not claim that Revolut has one global model, one global credit policy or one global data warehouse. It also does not claim that synthetic audit labels represent real protected-class prevalence. Instead, it shows how a governance monitor can enforce different data paths and evidence obligations while remaining comparable at the management-reporting level. This is why the prototype

separates the US processed dataset from the EU audit dataset and why the tool output uses a shared evidence schema without pooling sensitive attributes.

5. References

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